

sV2Pc: On Scaling LTE-based Vehicle-to-Pedestrian Communication using NDN

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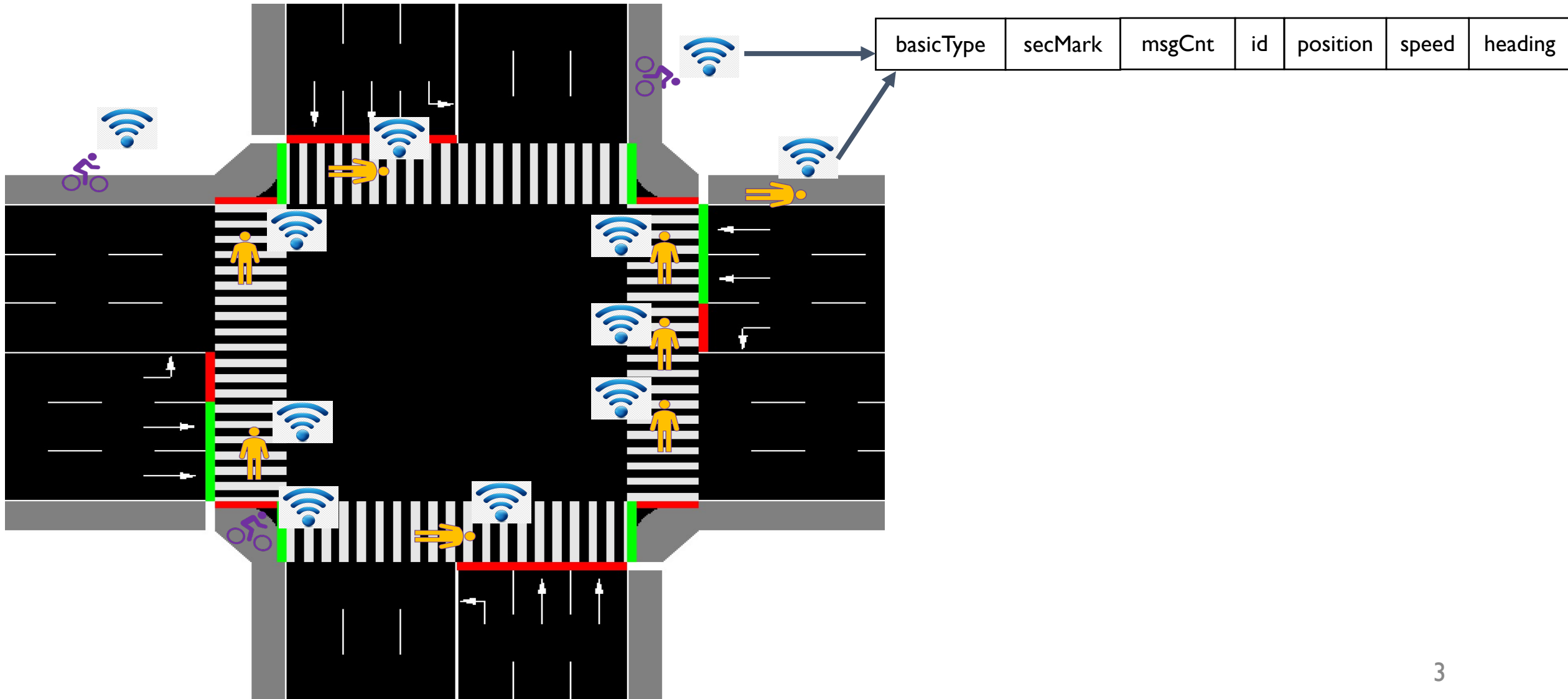
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Objectives

- Demonstrate advantages of Named Data Networking (NDN) in CV2X (LTE Direct) environment.
- Figure out several possible solutions to solve the scalability issues in vehicular environment in the context of a pedestrian safety application.
- Designed and developed an NDN forwarding strategy , named, **sV2Pc** suitable for vehicular environment to ensure a scalable communication system.

Personal Safety Message (PSM)



PSM in a crowded intersection



What if all these pedestrians in a crowded Intersection start sending PSMs in every second?

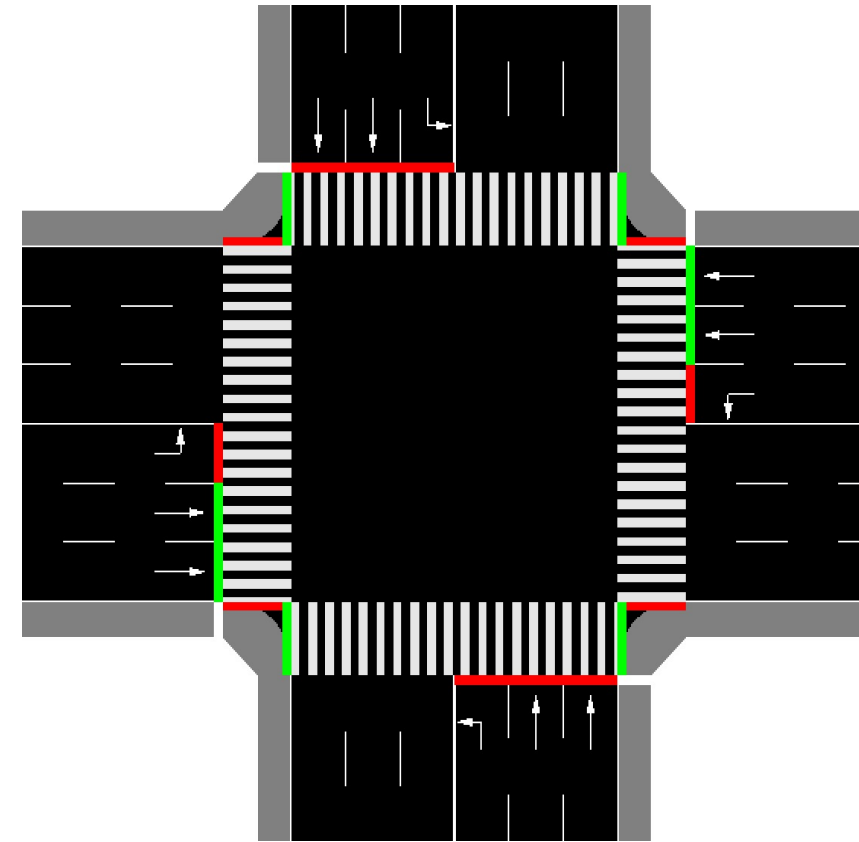
PSM broadcasts necessitate an additional bandwidth that may vary between **5 MHz up to 30 MHz** depending on pedestrian density.

sV2Pc

- To address this scalability issue in vehicular environment:
 - we identified advantages of data-centric communication of NDN, as well as challenges that need to be addressed to scale the communication.
 - we have used a simplified vehicle intersection awareness application, representing a class of V2P communications.
- We have designed a forwarding strategy **sV2Pc**, which follows a distributed and simple mechanism yet provides scalability by suppressing similar types of Interest and Data packets and geographical bounds for Interest forwarding.

Use case Application (NDN-PSA)

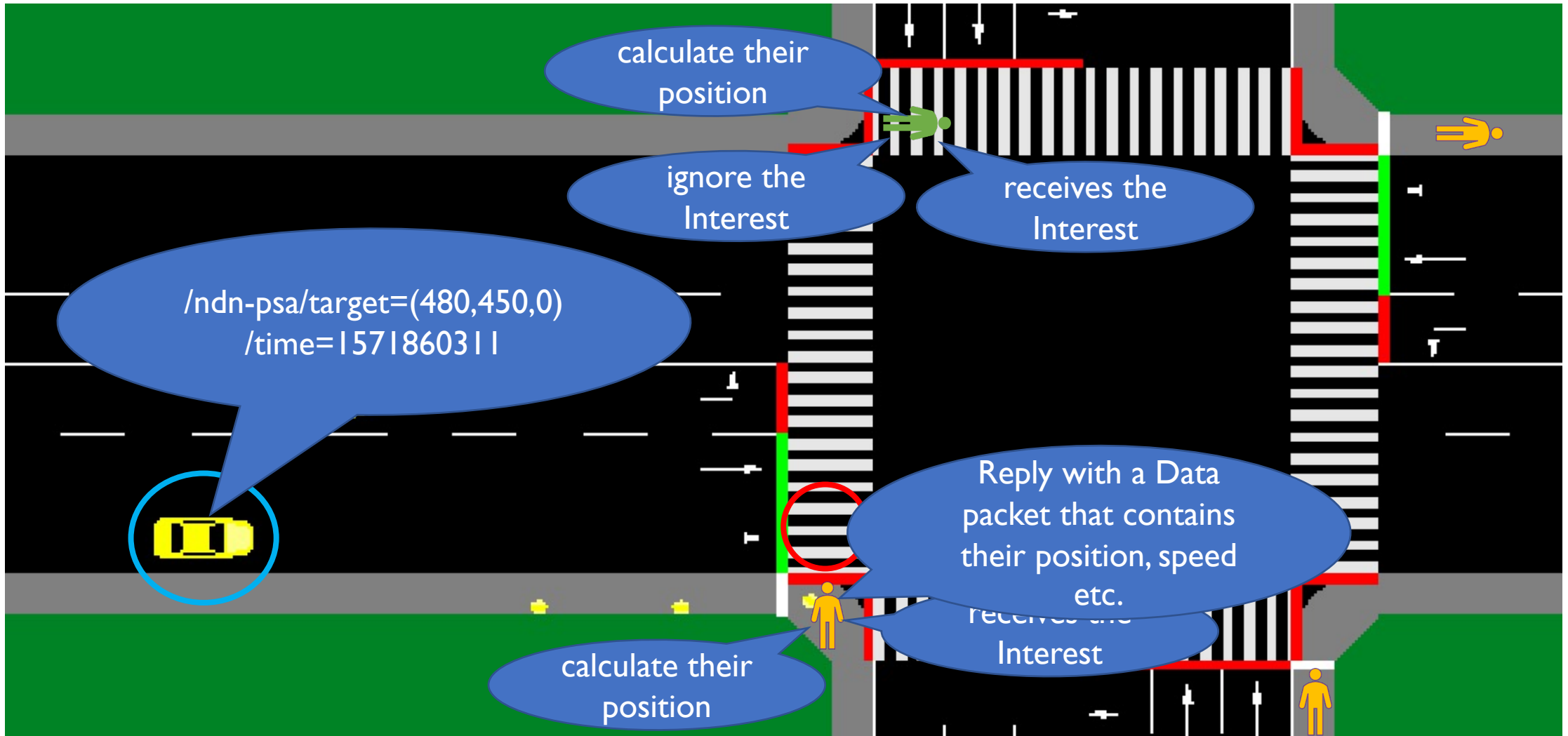
- To identify communication issues and highlight the proposed design decisions, we developed a simplified use case intersection awareness application named NDN-PSA.
- To be more specific, it will enable **communication** between **cars** (approaching an intersection) and **pedestrians** (crossing the intersection) using our request-response-based **NDN** mechanism to avoid any potential collision.



Naming Scheme

- “/**<app-name>**/**<region-of-interest>**/**<time>**”
- The “**region-of-interest**” will be a coarsely defined area at the intersection (for simulation, we have use coordinates) and the “**time**” component will represent the coarsely temporal dimension .
- However, these two components will be same for all vehicles that are querying about same region approximately in similar time bound that ensure the repeatability of Interest name.

Overview of NDN-PSA



Will any request-response based approach be feasible?



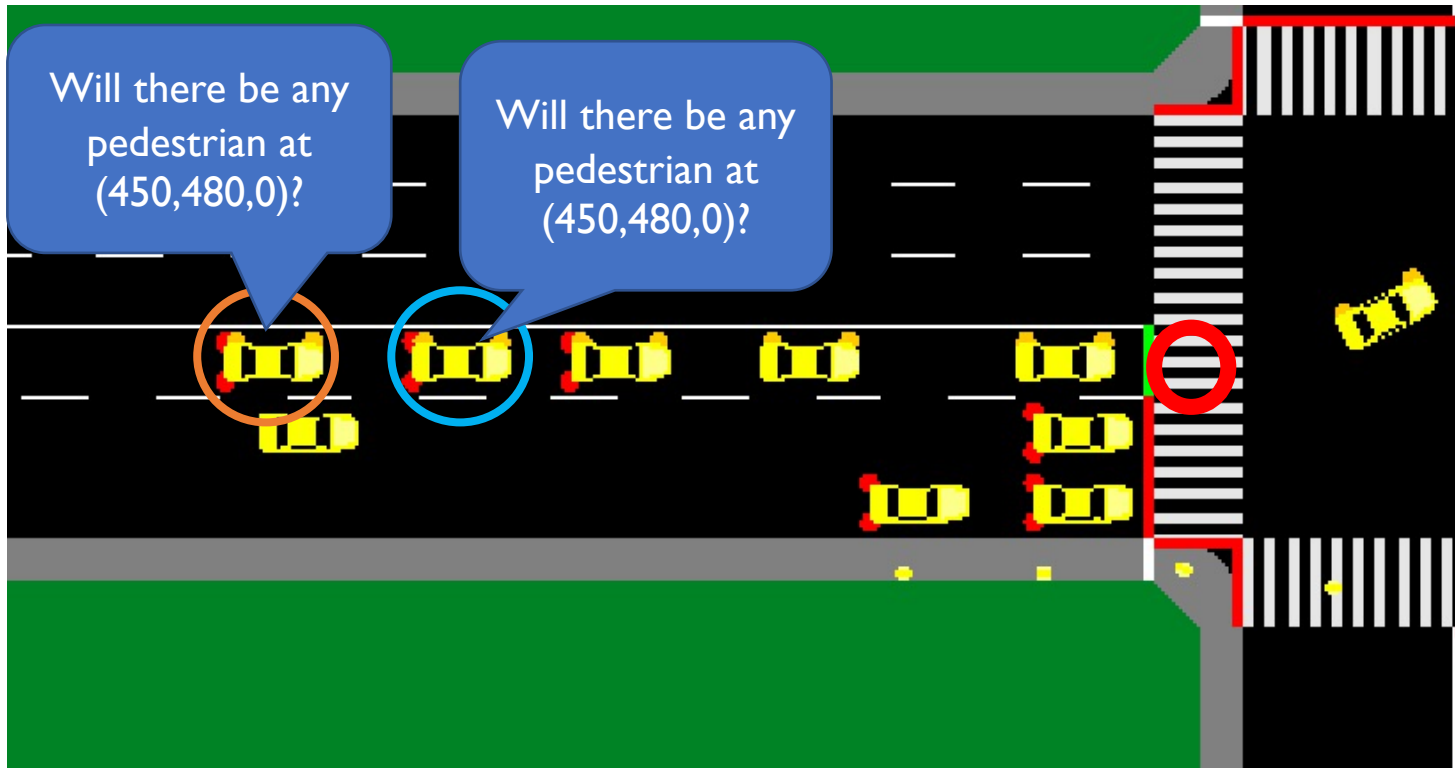
NAMED DATA NETWORKING

Here comes
Named Data
Networking
(NDN) with its
unique features to
provide a solution
for this issue and
make this
communication
feasible

What if in a busy intersection
with huge amount of traffic,
all the cars are trying to send
packets?

And again, in a crowded
intersection all the
pedestrians start replying?

Do all the cars need to ask question?



In a scenario, when both marked cars want to know about the same location, will it not be enough that just one car asked the question and another instead of asking the same question again, use the first car's answer?

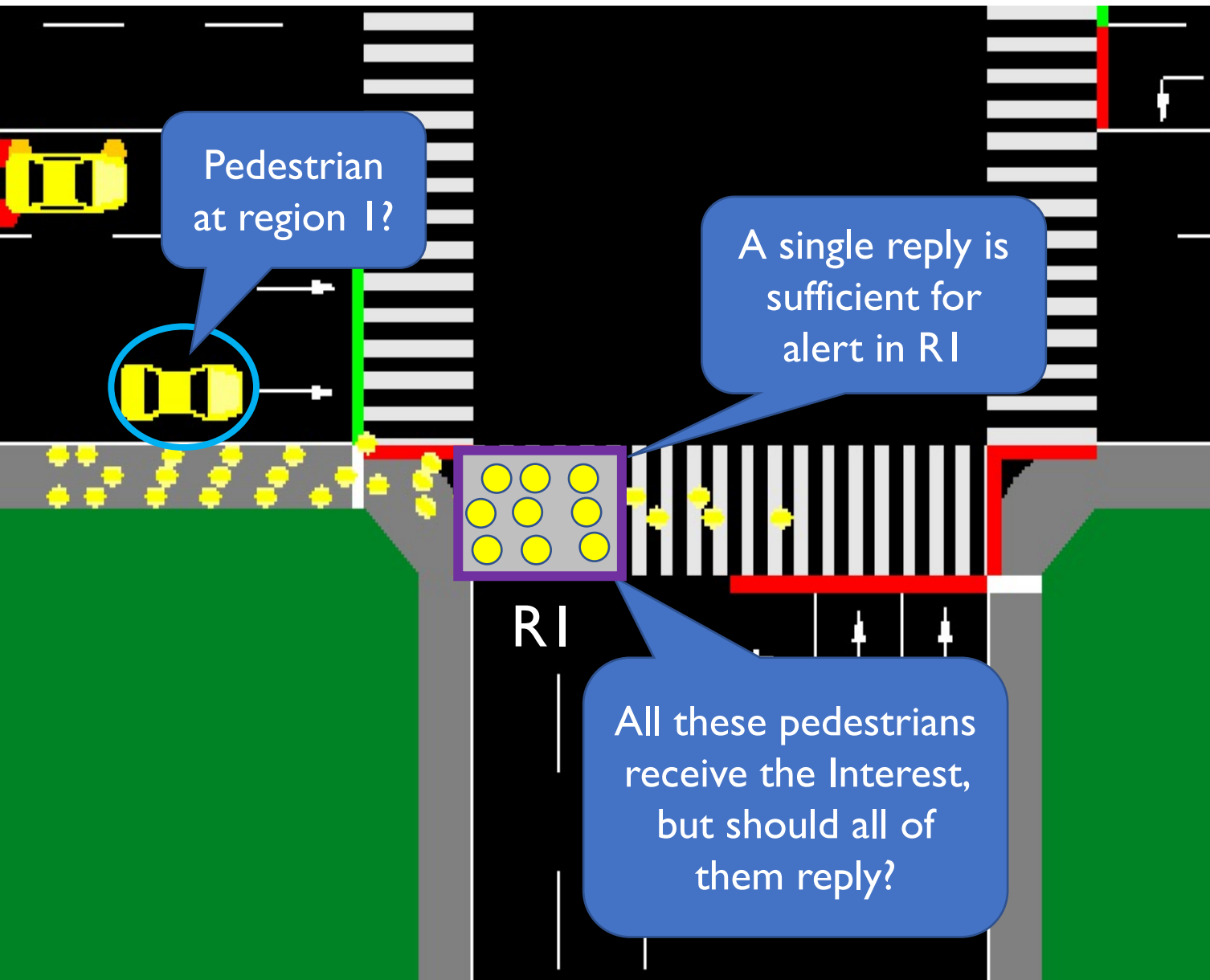
So, we have introduced a mechanism named **“Interest Suppression”** so that if a group of cars want to ask the same question, and one of them has already asked, then the others would suppress their own Interest.

What if all the pedestrians reply?



Again, for all the Interests being sent, if all the pedestrians reply with Data packets back, then again, we will generate a huge amount of traffic which is no better than PSM

Do all the pedestrians need to reply?

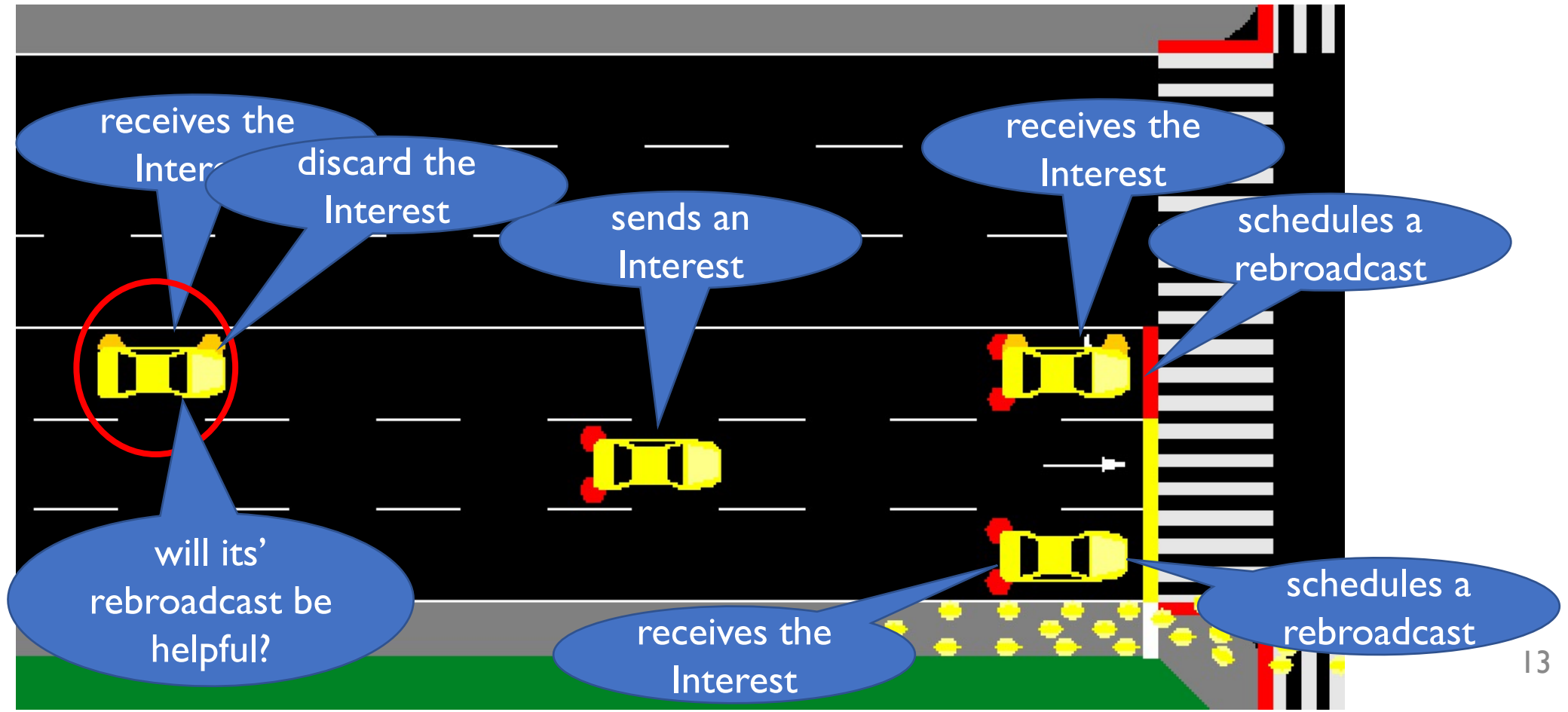


Based on this assumption, we have introduced “**Data Suppression**” mechanism to suppress unnecessary and redundant traffic.

In Data suppression, if any pedestrian receives an Interest, and it is about to reply, but some other pedestrian has already replied for the same Interest packet, then it would not reply any more.

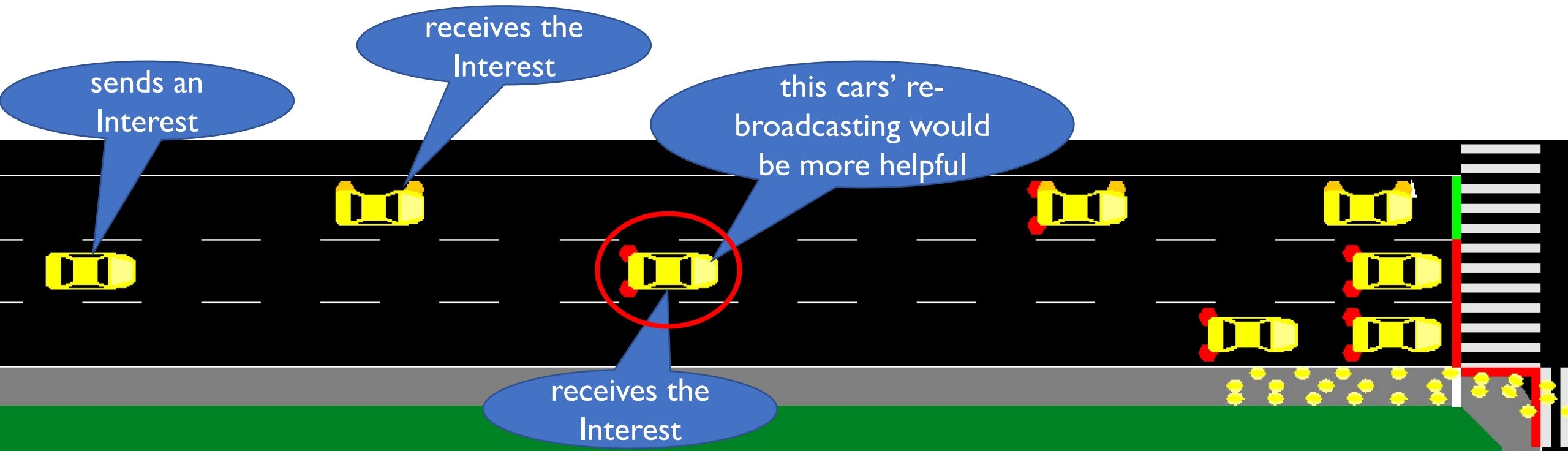
Directional Suppression

Forwarders should send Interests only in the direction of Data, considering the location of Data (from the name), its own, and Interest originator's locations.



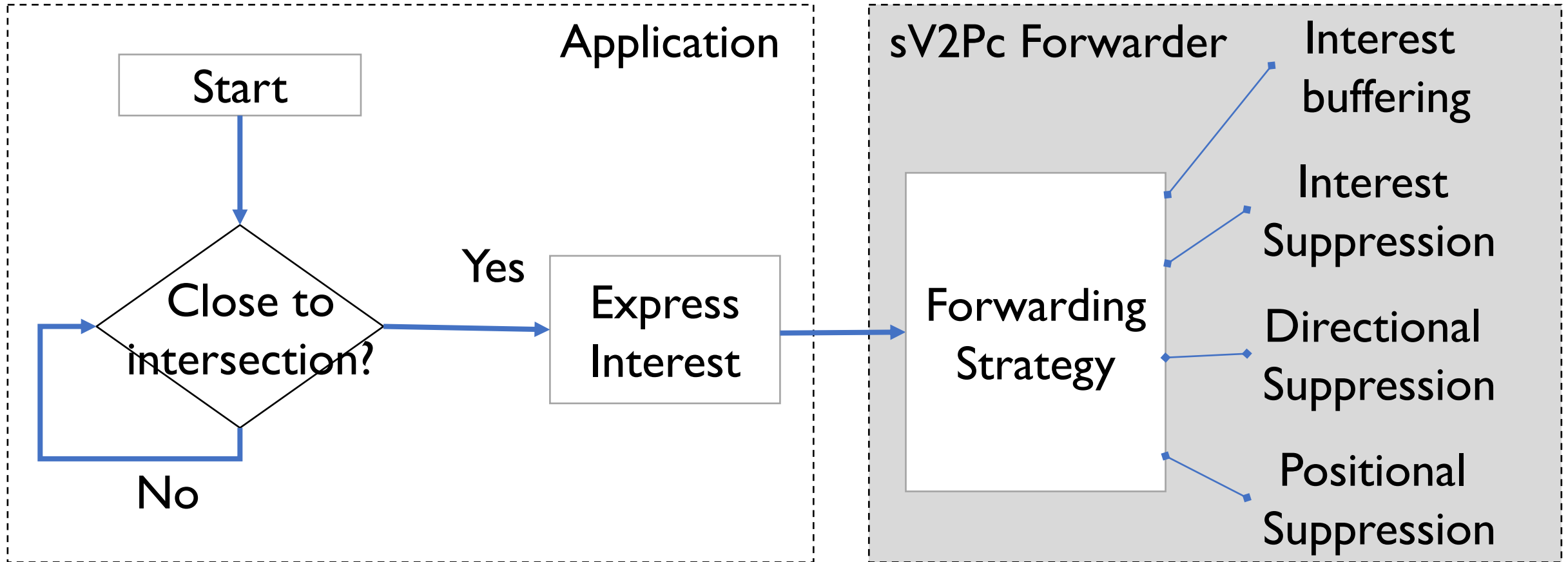
Positional Suppression

Interests should be re-broadcasted only by forwarders in the best position to do so.

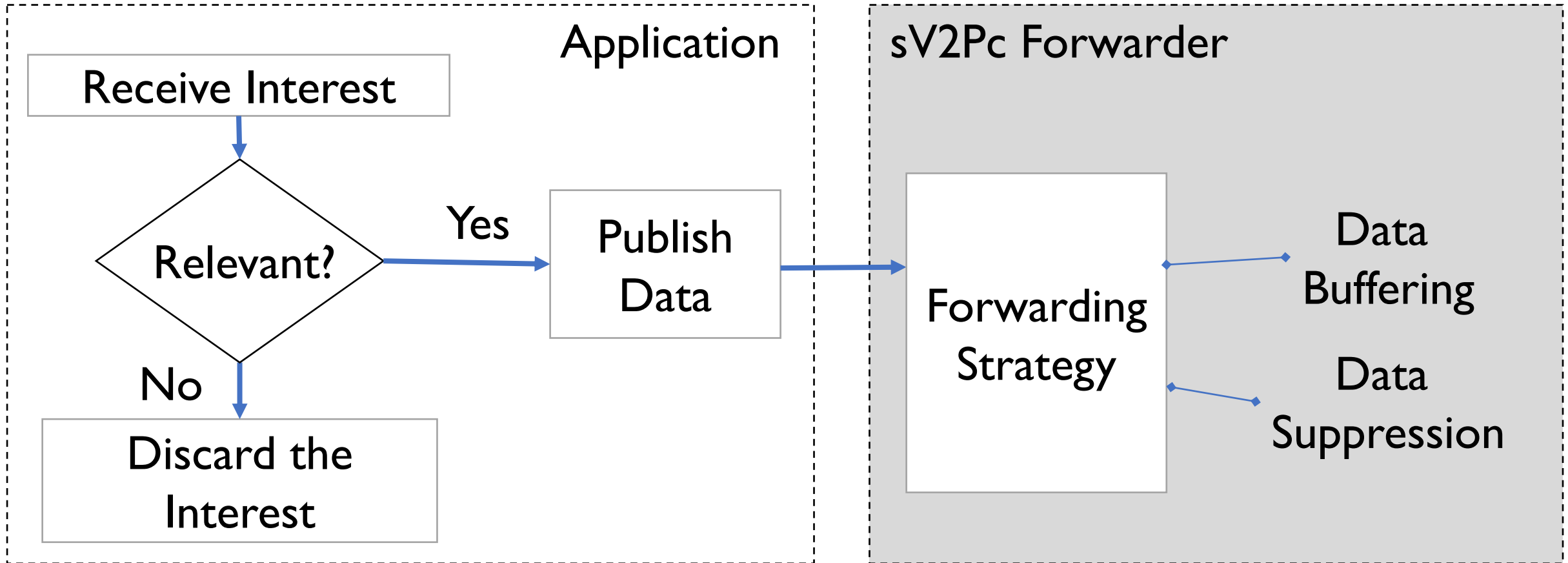


Ideally, rebroadcasting should happen by a car that is farthest from the originator of the Interest and closest to the place where the Data is.

From Cars' / Consumers' perspective

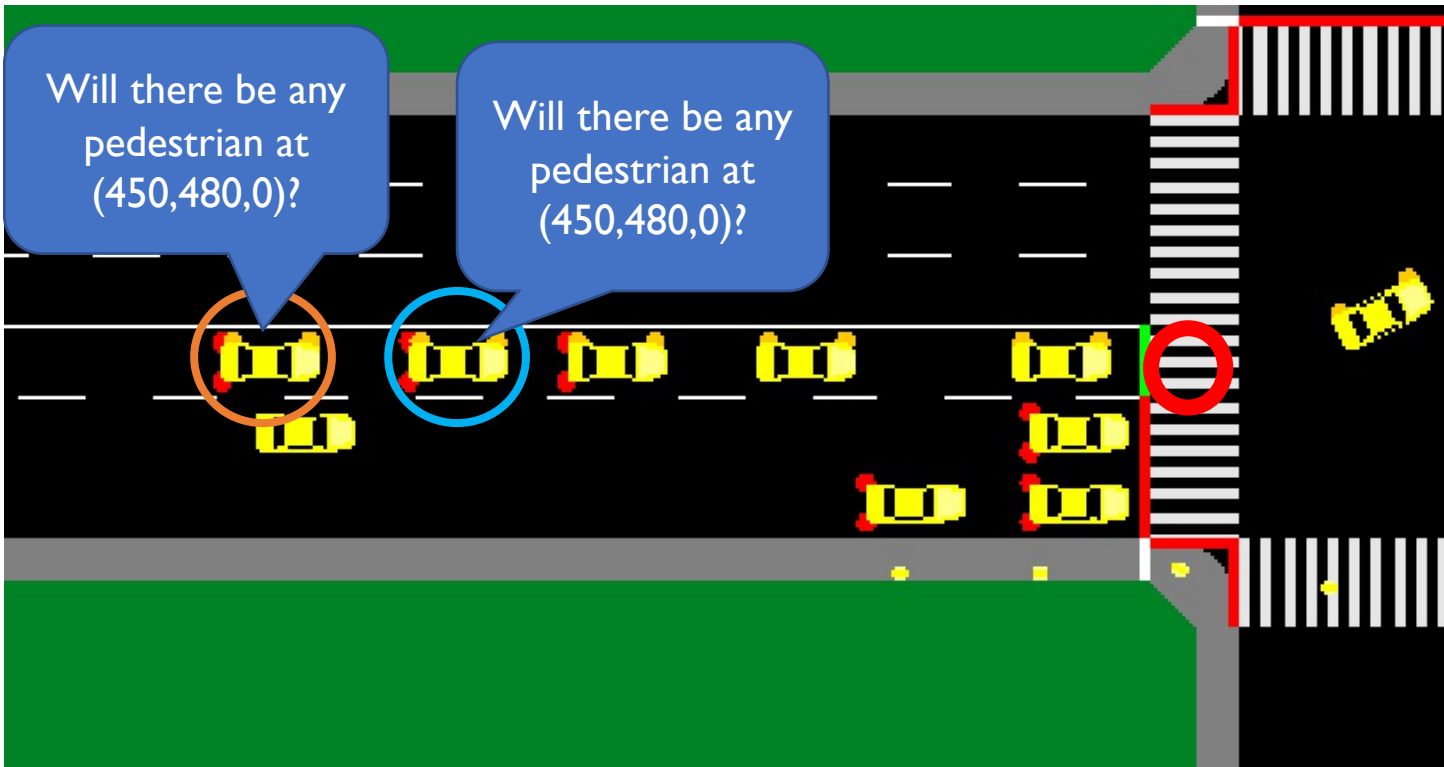


From Pedestrians'/Producers' perspective



How Interest Suppression Works?

- Before transmitting a new Interest, a node first checks if it receives the same Interest before.
- If all the cars send Interest at the same time (at the beginning of each second), then one would not hear others' Interests before transmission and no Interest suppression would happen.

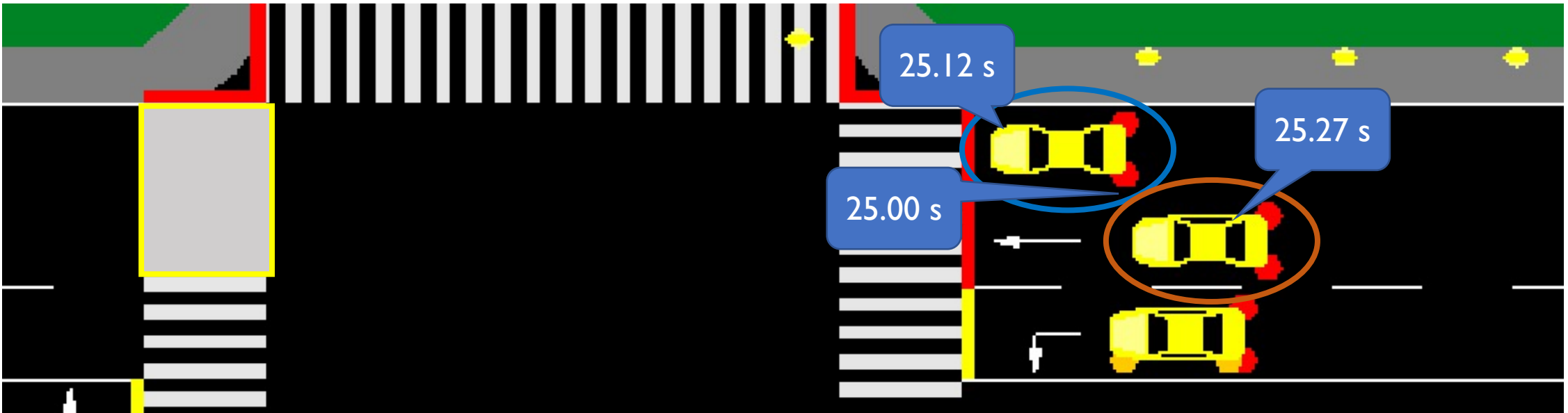
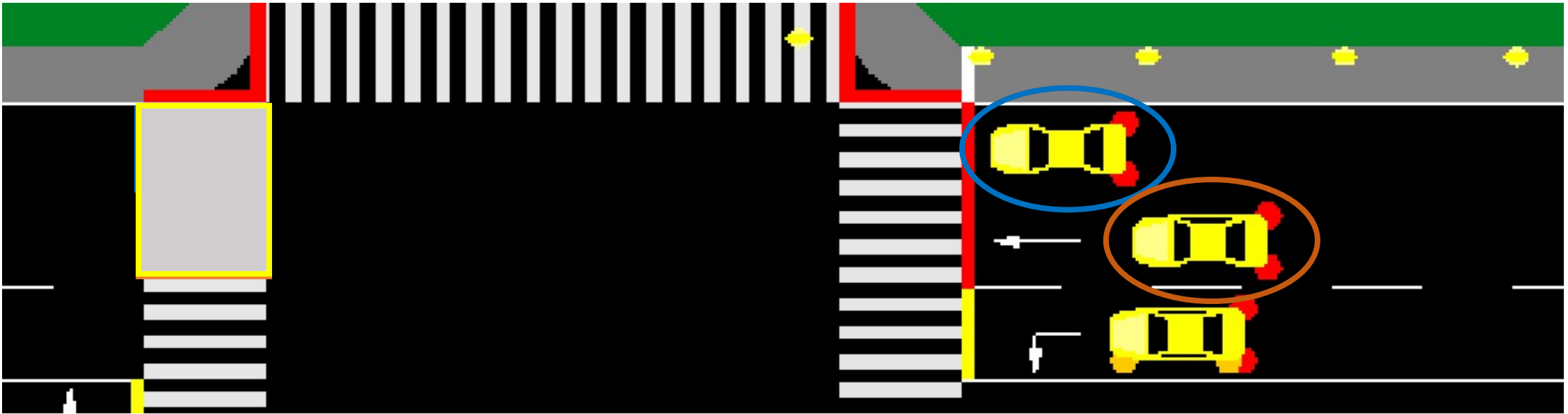


- So, we have added a random delay for the cars to transmit Interest so that cars who are scheduled to transmit later get a chance to hear the transmission of other cars that have transmitted earlier.

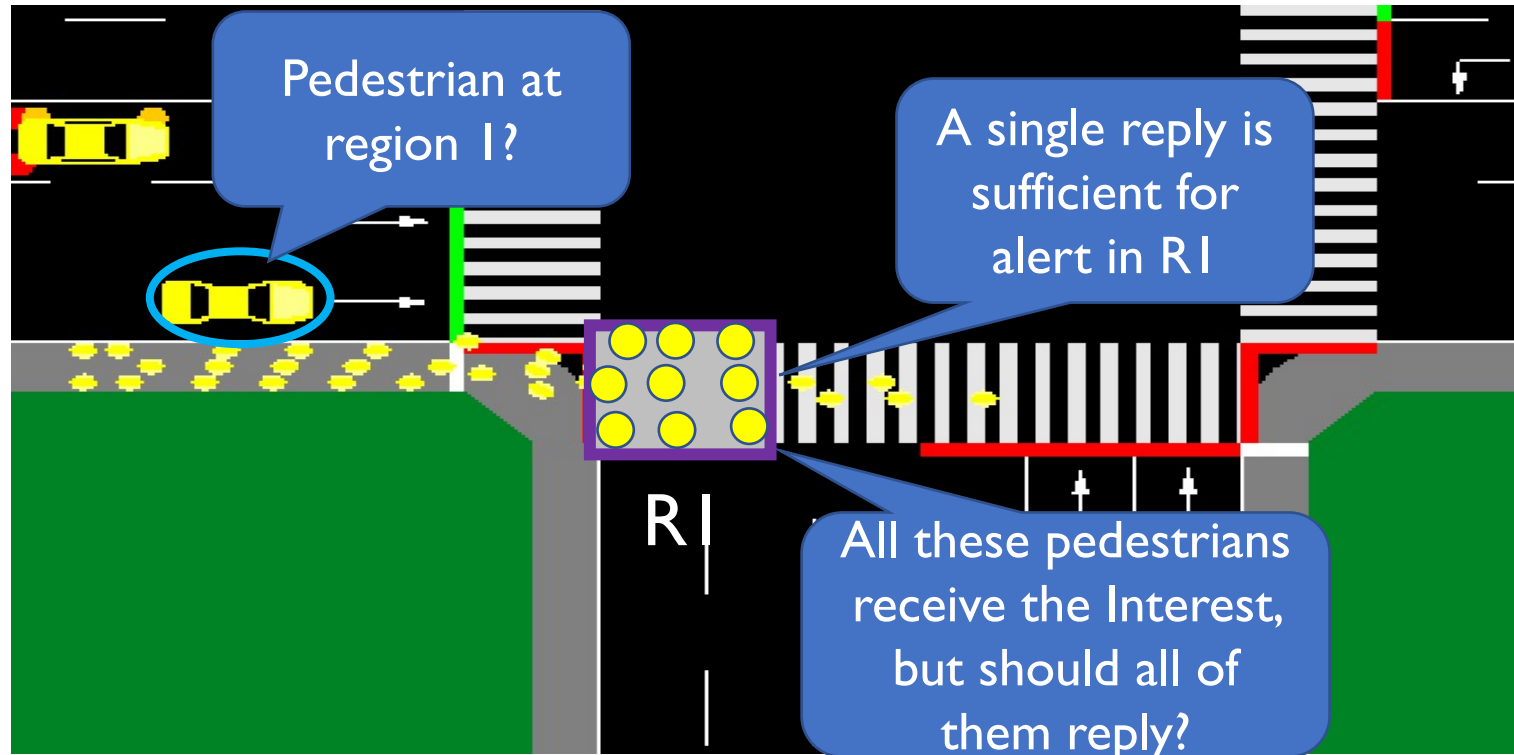
Condition to be the same Interest

- To be considered as the **same Interest**, the **Interest name** should be the same.
- A sample Interest name is following:
 - **/ndn-psa/target=(480,450,0)/time=1571860311**
- As application name is constant for our application, if **target** and **time** are same, then the Interest name would be the same.
- If the target location and time are very **granular**, then it would be rare that two Interest name would be same and so Interest suppression will not be occurred. So, we have made the location and time a bit coarse.

Making Target Location and Time Coarser



How Data Suppression works?



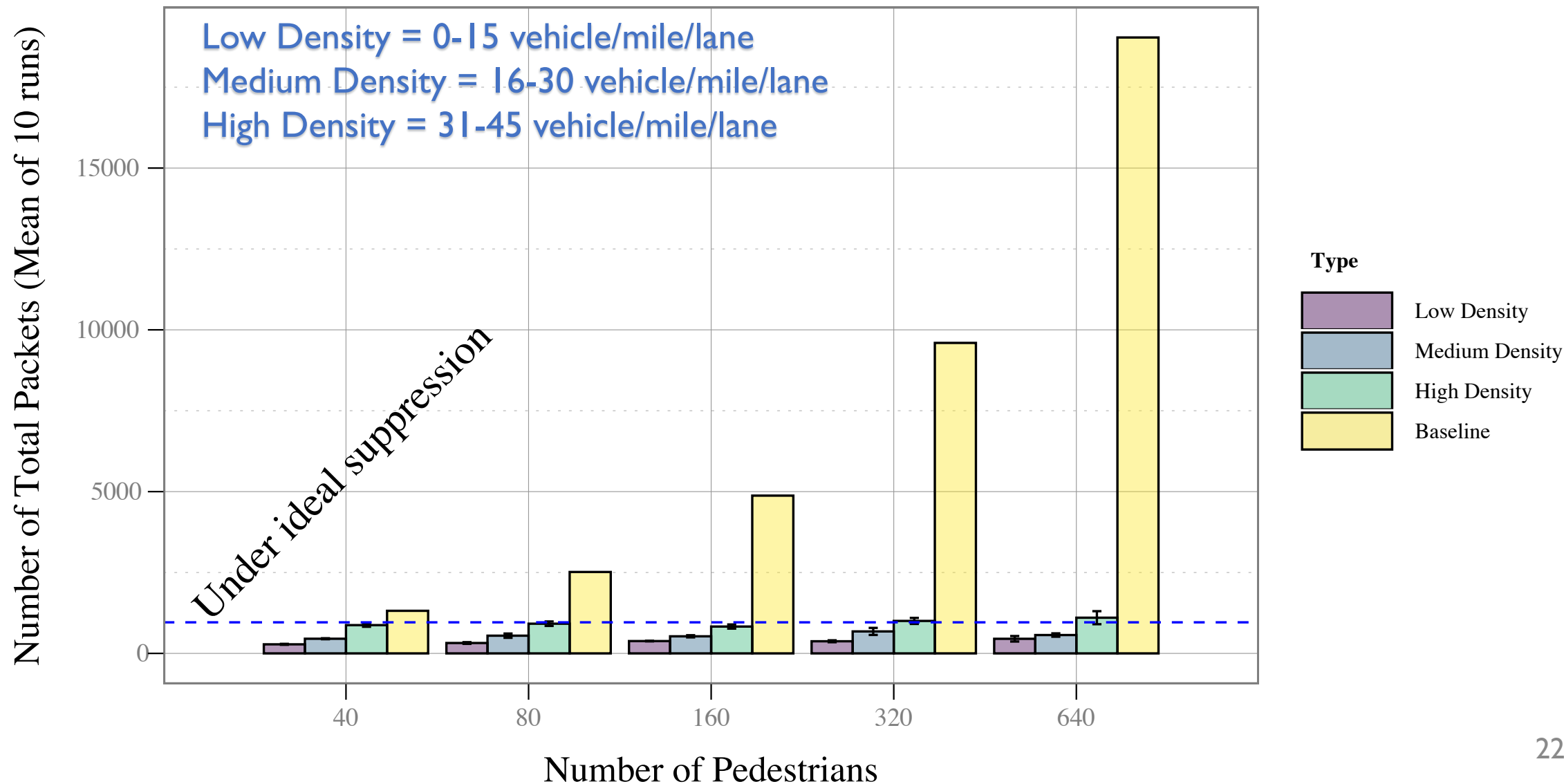
- Data Suppression works almost in the same way as Interest Suppression.
- Before sending a Data packet, a node checks if it receives any Data packet with the same Interest name before.

- Again, if all the pedestrian from a group of closely located people replies immediately after receiving an Interest, no suppression will ever happen.
- So, we have added a random delay to ensure that nodes that transmit later will get a chance for suppression.

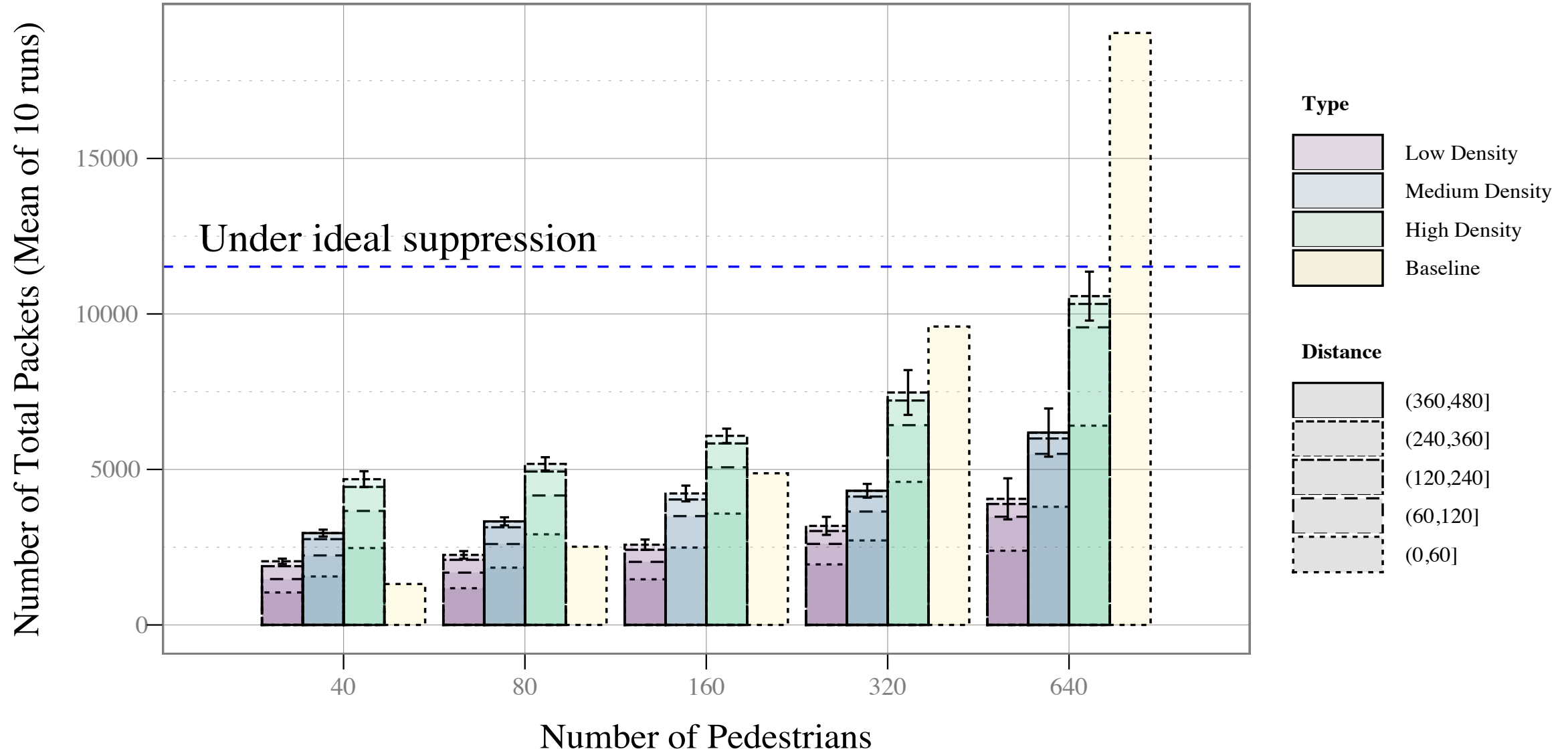
Evaluation

- To evaluate our application, we compare our application with PSM as baseline in terms of bandwidth for:
 - Different pedestrian count
 - Different vehicle density
 - Both single and multiple hops scenario.

Single Hop Scenario



Multiple Hops Scenario



Future Work

- Ensure proper security mechanism.

- Explore applicability of sV2Pc in different application scenarios.

- Improve suppression logic using centralized knowledge.

Thank you!

Q&A